When Should Adaptive Network Recovery Be Triggered

A Reproducible ns-3.47 Pilot on OLSR Resilience and Control Cost

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Research context

Fast routing control can shorten disruption, but periodic HELLO and topology messages consume wireless capacity and may destabilize service. Adaptive MANET routing and adaptive HELLO messaging have been studied previously (Qin and Kunz, 2008; Han and Lee, 2013). The practical question here is narrower: when should faster control be active, and at what cost?

Research question

Under stable, progressive, abrupt, and repeated disturbances, when does state-triggered OLSR timer adaptation improve packet delivery relative to static OLSR and always-on fast control, and what control and harmful-tail costs accompany any improvement?

Existing implementation

The ns-3.47 implementation uses a 16-node wireless ad hoc topology and two UDP flows. B0 uses standard OLSR timers. B1 keeps HELLO at 0.5 s and TC at 1.0 s. B2 applies the same fast timers after a signal or transmission-failure trigger. The v2 study added a repeated-disturbance condition and fixed the disturbed node across mechanisms for each seed.

Condition	Meaning	Runs per mechanism
A	Stable negative control	30
B	Progressive mobility disturbance	30
C	Abrupt next-hop radio failure	30
D	Repeated out-and-back mobility	30

Evidence and reproducibility

The package contains 360 fresh paired runs, per-run logs, raw and processed tables, deterministic keyed-bootstrap intervals, and checks for missing rows and disturbance-target mismatches. A clean ns-3.47 rebuild and full rerun in a separate directory reproduced the committed raw CSV byte for byte; regenerated tables and figures reported zero confidence-interval mismatches. This was an isolated rerun on the same host, not a third-party replication.

Selected sources. Qin L, Kunz T. Adaptive MANET Routing: A Case Study. ADHOC-NOW 2008. doi:10.1007/978-3-540-85209-4_4. Han SY, Lee D. IEEE Communications Letters. 2013;17(5):1040-1043. doi:10.1109/LCOMM.2013.040213.130076.

Verified findings

Condition	Always-on PDR effect	Triggered PDR effect	Always-on cost	Triggered cost
Stable	+0.571 [-0.630, +2.312]	+0.432 [+0.000, +1.296]	+0.05410	+0.00004
Progressive	+1.591 [+1.214, +1.971]	+0.108 [-0.069, +0.320]	+0.05305	+0.00307
Failure	+1.453 [+1.055, +1.796]	-0.117 [-0.279, +0.010]	+0.05043	+0.00111
Repeated	+3.392 [+2.736, +4.049]	+0.404 [-0.261, +1.055]	+0.05357	+0.00378

Reading the table. PDR effects are paired mean percentage-point differences from static OLSR with 95% percentile-bootstrap intervals. Cost is normalized OLSR control bytes per delivered payload byte.

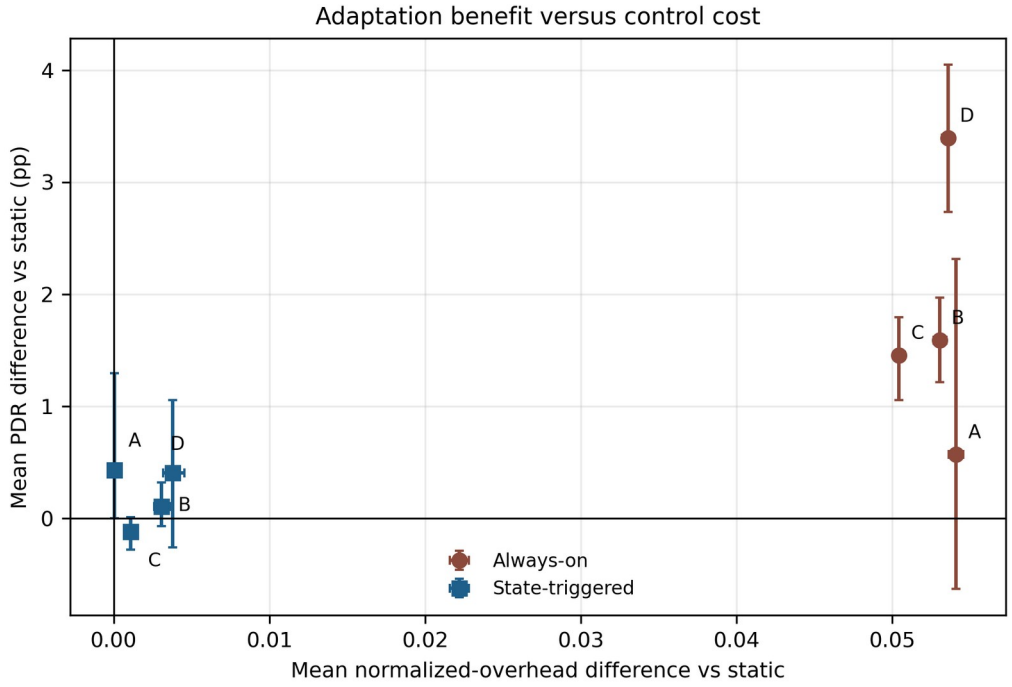


Figure 1 Mean delivery effect and normalized control cost relative to static OLSR. Error bars show 95% paired bootstrap intervals.

Interpretation and limitations

Always-on fast control improves mean delivery in all three disturbance conditions, but adds a large and consistent control cost and raises p95 delay in most runs. The locked triggered rule does not reproduce that benefit reliably. It also triggers in 3 of 30 stable runs, so the 20-event threshold is not validated. One topology family, two flows, deterministic path loss, and an aggregate recovery definition limit generalization.

Proposed doctoral extension

The next step is a guarded trigger and exit policy that combines route state with channel-load evidence, caps reactive duration, and treats worst-flow harm as a constraint. Confirmation requires fresh development and evaluation seeds, several topology and mobility families, heterogeneous traffic, and selected testbed checks.

Reproducibility and access

The published repository provides the locked protocol, source code, commands, raw and processed results, canonical statistics, figures, and clean-build reproduction record: github.com/wisam-makki-salim/adaptive-routing-resilience-ns3-pilot